

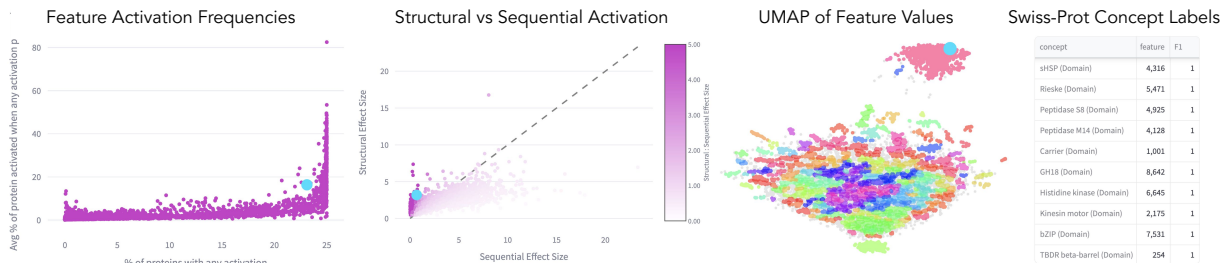
adjusting feature activation values steers language model outputs toward specific behaviors, demonstrating their causal role [20].

One of the field’s main challenges lies in evaluation. While technical metrics like how well autoencoders reconstruct their inputs are straightforward, assessing feature interpretability remains subjective. Recent work has made progress by using text-based games, where feature activations can be mapped to labeled game states in chess and Othello [21]. Protein studies offer similar potential through their structural and functional annotations, though interpreting biological features requires domain expertise that makes evaluation more challenging than in language or vision domains.

Our work with InterPLM demonstrates that while PLM neurons exhibit polysemantic behavior, SAE analysis reveals more interpretable features. We present a comprehensive framework of quantitative metrics and methods for visualizing, analyzing, describing, steering, and learning from these PLM-derived SAE features.

3 Results

a. Identify interesting features based on quantitative patterns



b. Visually explore feature activation variation across proteins

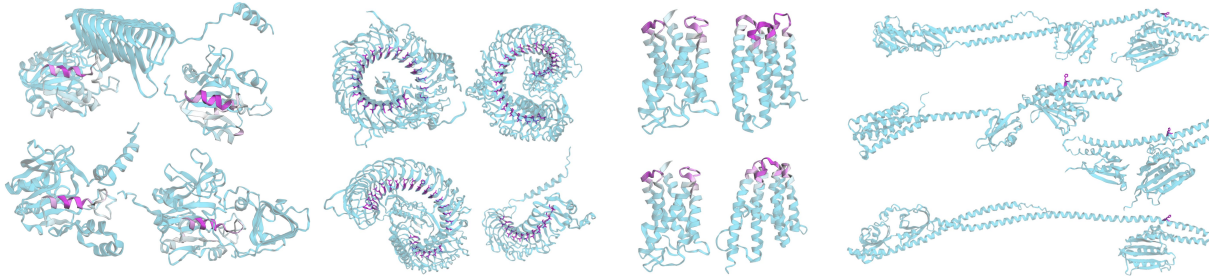


Figure 2: SAE feature analysis and visualizations reveal features with diverse and consistent activation patterns. a) Quantitative comparison of learned features through four complementary approaches: 1) Feature activation frequency distribution showing the relationship between proteome-wide prevalence (x-axis) and protein-specific activation strength (y-axis), revealing both ubiquitous and selective features 2) Structural vs. sequential activation patterns, comparing feature activation strengths in 3D versus sequence proximity to peak activation sites, revealing features that operate through either structural or sequential mechanisms 3) UMAP embedding of feature vectors, illustrating natural clustering of related structural/functional motifs 4) Swiss-Prot concept mapping results, linking learned features to known biological concepts. b) Structural visualization of four representative features (left to right: f/1854, f/10230, f/8144, f/8128) mapped onto example proteins, with each feature highlighted in the figure above it from (a) in blue.

3.1 Sparse Autoencoders Find Interpretable Concepts in Protein Language Models

To understand the internal representations of protein language models (PLMs), we trained sparse autoencoders (SAEs) on embeddings from ESM-2-8M. We expanded each of ESM-2’s six layers from 320 neurons to 10,420 latent features. The quantity and domain-specificity of these features necessitated developing both qualitative visualization methods and quantitative evaluation approaches to enable interpretation. See **Methods** for training and analysis details.

Analysis of feature activation patterns revealed three distinct modes of amino acid recognition: structural patterns (coordinated activation between spatially or sequentially proximal residues), protein-wide patterns (broad activation across single proteins), and functional patterns (consistent activation within annotated domains, motifs, and binding sites).

When evaluated with our quantitative measures of biological interpretability, we find significantly more conceptual features in SAE features than ESM neurons. We also show that this feature interpretation can be automated with a large language model, and once a feature has an interpretation, this can be used to suggest missing and new protein annotations.

Through clustering features based on their dictionary vectors, we identified groups that detect related biological structures with distinct specializations, from subtle variations in kinase binding site recognition to hierarchical relationships between specific and general beta barrel detectors. We further show that targeted interventions on specific feature values enable controlled manipulation of the model’s representations, producing interpretable changes that propagate to influence predictions even for non-intervened amino acids.

For consistency and clarity, our visualizations primarily focus on features from the fourth layer (denoted *f*/feature-number) unless explicitly stated otherwise.

3.2 InterPLM: An Interactive Dashboard for Feature Exploration

Features from SAEs of every ESM-2 layer can be explored through `InterPLM.ai`, which provides multiple complementary views of feature behavior: (1) sequential vs structural activation patterns, revealing how features capture both local sequence motifs and long-range structural relationships, (2) protein coverage analysis, distinguishing between features that identify specific local properties versus those capturing broader domain-level patterns, (3) feature similarity through UMAP [22] visualization, and (4) alignment with known Swiss-Prot [23] concepts. These feature’s characteristic activation modes often persist across diverse proteins, suggesting that many features capture consistent biochemical and structural properties (Figure 2). By interactively examining the full distribution of features through multiple lenses - their protein coverage, activation patterns, and spatial properties - we uncover a spectrum of behaviors, ranging from precise detectors that activate on single amino acids across many proteins to broader features that respond to complete sequences of specific protein families. As shown in Figure 1, examining the spatial distribution of activation sites uncovers features sensitive to various forms of proximity – whether sequential, structural, or both – providing insights into how the model encodes different aspects of protein organization.

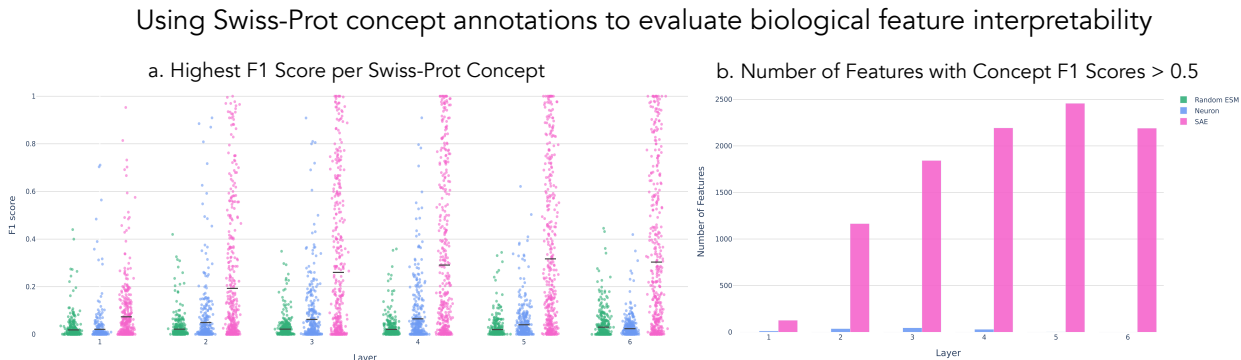


Figure 3: SAE features have stronger associations with Swiss-Prot concepts than ESM neurons. Comparing the features of an SAE model trained on ESM-2 embeddings (pink), the original neurons of the ESM-2 embeddings (blue), and the features of an SAE trained on embeddings from an ESM-2 model with shuffled weights (green). Models are compared based on the F1 scores between features and Swiss-Prot concepts. (a) For each concept, select one feature from the validation set (based on highest F1 with that concept), and visualize the F1 for that feature and concept in a held-out set. (b) For each layer, count the number of features that have an F1 score with any concept > 0.5 in both the validation and held-out sets.

3.3 Sparse Autoencoder Features Capture More Biological Annotations than Neurons

When comparing binarized feature activation values to biological concept annotations extracted from Swiss-Prot (dataset and methodological details in Methods), we observed that our learned features are much more specific than many concept annotations. The 433 Swiss-Prot concepts we evaluated span structural patterns, biophysical properties, binding sites, and sequential motifs (concept categories enumerated in Appendix C.2.1). While some concepts are specific to individual amino acids, others can cover broad domains that include many different physical regions, resulting in standard classification metrics overly penalizing features that activate on individual subsets of larger domains. For example, *f*/7653 activates on two conserved positions in tyrosine recombinase domains. While this feature has perfect

precision on the 45 tyrosine recombinases in our test set, it has a recall of 0.011 because tyrosine recombinase domains are hundreds of amino acids long so it 'misses' most example. To address this, we calculate precision of feature prediction per amino acid, but calculate recall *per domain*. In the case of the tyrosine recombinase feature, because it correctly identifies two amino acids in every domain tested, it now has a domain-adjusted recall of 1.0. Combining these to calculate the F1 scores between feature activations and concepts, we can quantitatively compare interpretability of SAE features versus the original neurons, even though features and concepts are defined at different levels of granularity.



Figure 4: Clustering reveals groups of features with similar functional and structural roles but subtle differences in activation patterns. (a) UMAP of SAE features clustered based on their dictionary values. Features associated with one of the top 20 most commonly labeled Swiss-Prot concepts are highlighted in color and severely random clusters (Swiss-Prot labeled or not) are manually identified. In particular, one cluster of kinase-binding-site features is expanded. (b) Structures of the maximum activating examples from 3 features selected from cluster of kinase-binding-site features. Higher activation values indicated with darker pink and location of catalytic loop specified. (c) Comparing the maximum activation value of each feature on the maximally activating proteins selected by the other kinases. (d) Comparing the physical locations within the kinase binding site these features activate, with specific annotations for each sub-region within kinase binding site that has feature activation. All features visualized on protein kinase A (Uniprot ID: P17612) (e) Structures of maximum activating examples from 3 features selected from cluster of TBDR beta barrels (f) Activation patterns of 3 proteins in a cluster unlabeled by Swiss-Prot concepts but identified as glycosyltransferase cluster. All features visualized on glycosyltransferase amsK (Uniprot ID: Q46638), the maximally activating protein for f/9047.

While individual neurons showed at most 46 clear concept associations per layer, SAE features revealed up to 2,548 features per layer with strong concept alignment (Figure 3). This difference is explained by both the SAE features' ability to capture more specific qualities within each concept category and their detection of a broader range of concepts overall, expanding from 15 distinct concepts with neurons to 143 with SAE features. Full list of associated Swiss-Prot